

Mr Satwik Cheppala

Role: Senior Data Scientist
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Profile

Senior Data Scientist with 5+ years of experience designing, deploying, and scaling machine learning, forecasting, and optimization solutions across marketplace, fintech, and on-demand logistics platforms. Expertise spans demand and supply forecasting, dynamic pricing, fraud detection, workforce and inventory optimization, ETA prediction, dispatch systems, and last-mile logistics optimization. Demonstrated success delivering production-grade ML systems at Bolt, Tide, Zepto, and Uber, driving significant improvements in conversion, operational efficiency, fulfillment performance, cost optimization, and service reliability at scale. Strong technical foundation in Python, SQL, PySpark, Airflow, MLflow, OR-Tools, and modern cloud-based data platforms.

Experience

05/2026 – present
Tallinn, Estonia

Senior Data Scientist - Pricing Bolt Technology oü

- **Re-architected** driver-queue feature handling in the Offer Dynamic Pricing (ODP) conversion model (XGBoost, 5+ markets) after tracing **+39pp** → **~14pp city-level calibration** bias to train/serve skew in features driving **~51% of importance** — replacing per-city patches with a single representation fix.
- **Shipped** a city-gated hinge correction to the S20 price-elasticity engine, cutting out-of-sample error **~40%** (**~5pp** → **~2.5pp**) in gated cities; caught the underlying **~5pp conversion error at +50%** by validating Tanh extrapolation against **+20%/+50% arms**, and gated rollout behind a 3-gate rule.
- **Built** a discount-budget regularizer (λ_{disc}) with per-city dual-ascent tuning as a symmetric GMV brake, after surfacing production defects (serve-time outlier clipping, priority-request crashes) silently degrading **2–13%** of traffic per city.

01/2026 – 04/2026
Hyderabad, India

Data Scientist 3 Tide Financial Technologies Private Limited

- Shipped Tide's production acquiring-fraud stack — **28-rule engine + v11 XGBoost model** under a unified **1,000 alerts/month** budget — capturing **~40% of fraud £-value** on a 2026 out-of-time hold-out at flat ops workload; pinpointed **PAYLINK** as the dominant channel, isolating **55% of chargeback £-value in 3% of volume**.
- Reframed the objective from AUC to **value-recall-at-capacity** and shipped a **250 → 10,000 alerts/month tradeoff table (40+ operating points)**, letting ops size review headcount against marginal £-recall and select a capacity tier directly.
- Deployed **ProductionAlertGate**, a **3-gate** inference pipeline that narrows **~3M candidate transactions** → **1,000 alerts/month** (budget cap → quantile-derived amount floor → rule trigger), validated via chronological **12-month replay** to eliminate look-ahead and match live review limits.
- Engineered **60+ point-in-time features** across **4 rolling windows (7/30/60/90d)** + Tecton signals; caught and fixed **bin-total leakage in 3 legacy features**, then hardened selection with **3-fold time-series CV (90-day embargo)** and **50-trial Optuna HPO over 9 hyperparameters** scored directly on value-recall-at-cap.

07/2024 – 12/2025
Bangalore, India

Senior Data Scientist Zepto (YCombinator 20)

- Designed personalized re-ranking blending real-time user, cohort, and live-availability signals (FAISS ANN retrieval + gradient-boosted re-ranker), **driving 8–12% uplift in search-led GMV** while holding p95 latency under 100 ms.
- Built a learning-to-rank search engine (LightGBM LambdaMART, migrated to a two-tower neural retriever) for product discovery, lifting **search-to-cart conversion by 14–20%**, cutting zero-result queries by 35%, and improving **NDCG@10 by ~12%**.
- Developed AutoML and probabilistic forecasting—powered safety-stock & replenishment systems using PySpark, SQL, and Airflow, **decreasing stockouts by 20%** and **reducing excess inventory by 15%**.
- Automated network load-balancing using PySpark + AWS Lambda, **improving throughput by 20%**.
- Developed a reinforcement learning—based decision engine (PPO/DQN) to **optimize real-time discount and incentive decisions**, modeling short-term demand elasticity and **improving margin by 12–18%**.

11/2022 – 07/2024
Hyderabad, India

Senior Data Scientist Uber

- Built the **ML-driven Retriever model** using **Gradient Boosting + Ensemble Classification**, automating driver reactivation workflows and delivering an **86% reduction in Reactivation TAT**, **16% reduction in contact inflows**, and **\$350–500K annual arrears savings**; deployed to **35% of total supply** in the India South Asia region.
- Applied **ML-driven analytics** (EDA, cohort & funnel analysis, uplift modeling, cost optimization) using **Gradient Boosting, ExtraTrees, and Logistic Regression** to uncover supply-funnel drivers, increasing **First Trip conversions by 12%** and improving **activation efficiency by 18%**.
- Built predictive churn and re-engagement models for driver supply using ensemble ML techniques that

Experience

improved weekly active **driver retention by 8–12%** and reduced avoidable driver **churn by ~15%**.

08/2021 – 11/2022
Pune, India

Data Scientist IQVIA

- Built **ML pipelines on 45M+ US patient claims** using **Python and AWS Athena**—covering **feature engineering, cohort creation, and model training** (GBM, Logistic Regression, K-Means)—to **predict treatment drop-offs, classify patient pathways, and model physician prescribing patterns** for NMIBC.
- Built **predictive physician targeting models** using **Propensity Scoring, GBMs, and Uplift Modeling**, applying them to large-scale patient and prescribing datasets to identify **high-value physicians** and driving a **37% QoQ increase in sales**.
- Executed advanced Sales & Marketing analytics—including market landscaping, salesforce sizing, HCP alignment, segmentation and targeting, patient profiling, journey assessment, forecasting, incentive compensation, and dashboarding—**leveraging ML-enhanced modeling to improve targeting accuracy by approximately 20%**.

Skills

– LANGUAGES

Python
Professional

SQL
Professional

PySpark
Professional

– MACHINE LEARNING

Scikit-learn
Professional

XGBoost
Professional

Numpy,Scipy,Pandas
Professional

Deep Learning
Professional

Pytorch,TensorFlow
Professional

Statsmodels,statistics
Professional

– GENERATIVE AI & LLMS

OpenAI API
Professional

LangChain
Professional

HuggingFace Transformers
Professional

– DATA ENGINEERING

Airflow
Professional

Spark
Professional

AWS Glue
Professional

Snowflake
Professional

– MLOPS

MLflow
Professional

Docker
Professional

Kubernetes
Professional

AWS Sagemaker
Professional

– VISUALIZATION

Tableau
Professional

Matplotlib,Seaborn
Professional

Plotly
Professional

– REINFORCEMENT LEARNING & OPTIMIZATION

Constrained PPO
Professional

Contextual bandits
Professional

Lagrangian dual ascen
Professional

– CAUSAL INFERENCE & EXPERIMENTATION

Switchback design
Professional

CUPED
Professional

Uplift Modeling
Professional

A/B testing
Professional

– CLOUD

GCP,AWS
Professional

AZURE
Limited

Education

08/2017 – 05/2021
Rourkela

Electrical Engineering | Bachelor of Technology
National Institute of Technology, Rourkela